

Pushpa Sharma

AI/ML Engineer • Gen AI • Agentic AI • LLM Systems

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SUMMARY

AI Engineer with 9 years of software engineering experience, including 3+ years building production Gen AI, Agentic AI, and ML applications for banking, automotive, and healthcare clients. Designs and ships end-to-end AI solutions — RAG pipelines, LLM integrations, multi-agent systems, and conversational AI — underpinned by 6 years of full-stack product engineering across banking, healthcare, media, and security domains. Proficient in Python, LangChain/LangGraph, TensorFlow, PyTorch, and vector databases, with scalable deployment via FastAPI, Docker, and Kubernetes.

CORE SKILLS

Gen AI & Agents	LLM Integration (GPT, Claude, Gemini, Llama)	Agentic AI	RAG & Advanced RAG
	Prompt Engineering	Fine-Tuning	Function Calling
		MCP	Semantic Search
Frameworks	LangChain	LangGraph	LangSmith
	LlamaIndex	AutoGen	CrewAI
	Hugging Face	TensorFlow	PyTorch
	Scikit-learn	XGBoost	
Data & Infra	Pinecone	FAISS	Chroma
	Weaviate	PostgreSQL	NumPy / Pandas
	Azure OpenAI	AWS Bedrock	Databricks
	Ollama		
Engineering	Python	FastAPI	Flask
	LLMOps / MLOps	CI/CD for LLMs	Model Versioning
	Docker / Kubernetes	Git	React.js / Next.js

EXPERIENCE

AI/ML Engineer — TEKsystems (NatWest Bank) 2025 - Present • Bengaluru

Agent Toolkit AI — Gen AI modernization of the bank-agent platform

- Modernized the legacy Bank Agent Toolkit into an AI-powered application; built RAG pipelines to securely retrieve customer information from internal databases and knowledge sources.
- Developed AI-powered assistants for income analysis, planning recommendations, and direct-debit setup; built prompt templates and chaining logic with LangChain and LangGraph.
- Enabled semantic search of financial records with FAISS/vector databases; integrated LLM inference and embeddings via Azure OpenAI and FastAPI.

Ask Archie — Gen AI employee-support assistant

- Built an LLM-based assistant (OpenAI GPT, LangChain, Pinecone/Weaviate) answering workplace queries from company policy docs, FAQs, and the knowledge base via RAG.
- Designed confidence-based fallback and escalation: low-confidence or dissatisfied users are auto-transferred to live human support over WebSocket/SignalR.
- Exposed real-time LLM inference and ML models as REST APIs with FastAPI; designed and developed the employee-facing chatbot UI.

Senior Software Engineer, AI — Grid Dynamics (Ford) 03/2024 - 01/2025 • Bengaluru

Agentic AI vehicle-exploration chatbot

- Built an Agentic AI chatbot (GPT-4, Claude, LLaMA with LangChain/LangGraph) that helps customers explore Ford vehicles — design imagery, specs, pricing, and availability.

- Enabled agent tool-calling against external APIs (availability, pricing, dealership information) via an MCP server; implemented a RAG pipeline over Ford's vehicle database using Pinecone.
- Optimized models using ensemble techniques, reducing computation time by 40% while maintaining model performance; exposed AI/ML models as FastAPI services.
- Built the conversational React.js interface and collaborated with UX designers to integrate AI seamlessly into the product experience.

Senior Software Engineer — Emids (Humana)

08/2022 – 04/2024 · Bengaluru

- Integrated an AI system into the Solara single-sign-on portal for medical employees; designed AI algorithms that improved predictive capability on client data (PyTorch, TensorFlow, Hugging Face, vector DBs).
- Applied prompt engineering to early Gen AI features; developed reusable components in Next.js and React.js and built supporting services in Node.js and Python.

Advisory System Analyst — IBM

01/2022 – 07/2022 · Bengaluru

- Implemented ML/DL models in production (TensorFlow, PyTorch, Scikit-learn, XGBoost) with data processing in NumPy/Pandas, served through FastAPI and Flask.
- Reduced API errors through Redux-Saga state management and built the React.js data-management UI over PostgreSQL/NoSQL stores.

UI Stack Developer — Innominds

03/2020 – 01/2022 · Bengaluru

- Developed the Vector Flow physical-security platform UI (React.js, Redux, Syncfusion), building reusable static and dynamic components for data-dense operational dashboards.

Software Engineer — AH Infotech (Sony Music)

05/2017 – 03/2020 · Bengaluru

- Delivered the Artist 360 media and entertainment platform end to end, including full CRUD functionality (React.js, Node.js, Express, SQL, REST APIs).

— EDUCATION

Diploma, Data Science & AI

IIIT Bangalore · 2025 – 2026

B.Tech

Biju Patnaik University of Technology · 2012 – 2016

— KEY ACHIEVEMENTS

- Reduced model computation time by 40% through ensemble optimization while maintaining accuracy.
- Increased user engagement by 20% by delivering impactful, AI-powered product features.